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ThyroNet-XAI: An Explainable Multimodal Framework for Thyroid Cancer Analysis Using Clinical and Ultrasound Data

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Abstract

ThyroNet-XAI integrates clinical tabular data and thyroid ultrasound images for explainable thyroid analysis.

- Clinical branch: SMOTE-ENN balancing, feature selection, and hybrid Random Forest–DNN classification.
- Imaging branch: FCN–TransUNet for thyroid nodule segmentation.
- Explainability: SHAP and LIME for clinical predictions; Grad-CAM for image-based interpretation.
- Evaluation: Kaggle thyroid clinical data, TN3K segmentation data, and DDTI external evaluation.

Clinical branch

6,916 clinical records

Ultrasound segmentation

TN3K: 3,493 images

External evaluation

DDTI: 637 images

Goal: combine predictive performance with transparent, multimodal analysis.

Introduction

Clinical information

Biochemical and demographic features can support risk assessment, but data may be imbalanced.

Ultrasound imaging

Images provide spatial evidence for thyroid nodule analysis and segmentation.

Interpretability

High-performing AI needs understandable outputs and transparent evidence.

Key gap

Many approaches address clinical prediction or ultrasound analysis separately. Class imbalance and limited interpretability further constrain practical use. ThyroNet-XAI combines these capabilities in a single explainable framework.

Motivation & Objectives

MOTIVATION

- ➢ Clinical data are severely imbalanced.
- ➢ Ultrasound segmentation needs both local detail and global context.
- ➢ Clinical and image predictions should be interpretable.
- ➢ Generalization should be examined on an independent dataset.

OBJECTIVES

- ❑ Build a multimodal **clinical** + ultrasound framework.
- ❑ Improve minority-class handling using SMOTE-ENN.
- ❑ Combine RF-DNN classification and FCN-TransUNet segmentation.
- ❑ Provide SHAP, LIME, and Grad-CAM explanations.
- ❑ Evaluate clinical performance and cross-dataset segmentation.

RESEARCH QUESTION

TBA

Literature Review

Research direction	Strength	Common limitation
Clinical ML / ensembles	Uses biochemical and demographic features	Often tabular-only; imbalance remains challenging
CNN / Transformer imaging	Learns image features and spatial context	May omit clinical context
Multimodal methods	Combines complementary information	Fusion and external evaluation vary
Explainable AI	SHAP/LIME and heatmaps aid interpretation	Often applied to separate branches

Research gap addressed

A unified pipeline for imbalance-aware clinical prediction, ultrasound segmentation, multimodal interpretation, and external evaluation.

Datasets

Dataset	Modality	Size	Role
Kaggle thyroid dataset	Clinical/tabular	6,916 records	Clinical classification
TN3K	Ultrasound images	3,493 images	Training / internal segmentation evaluation
DDTI	Ultrasound images	637 images	Independent external evaluation

Clinical preprocessing

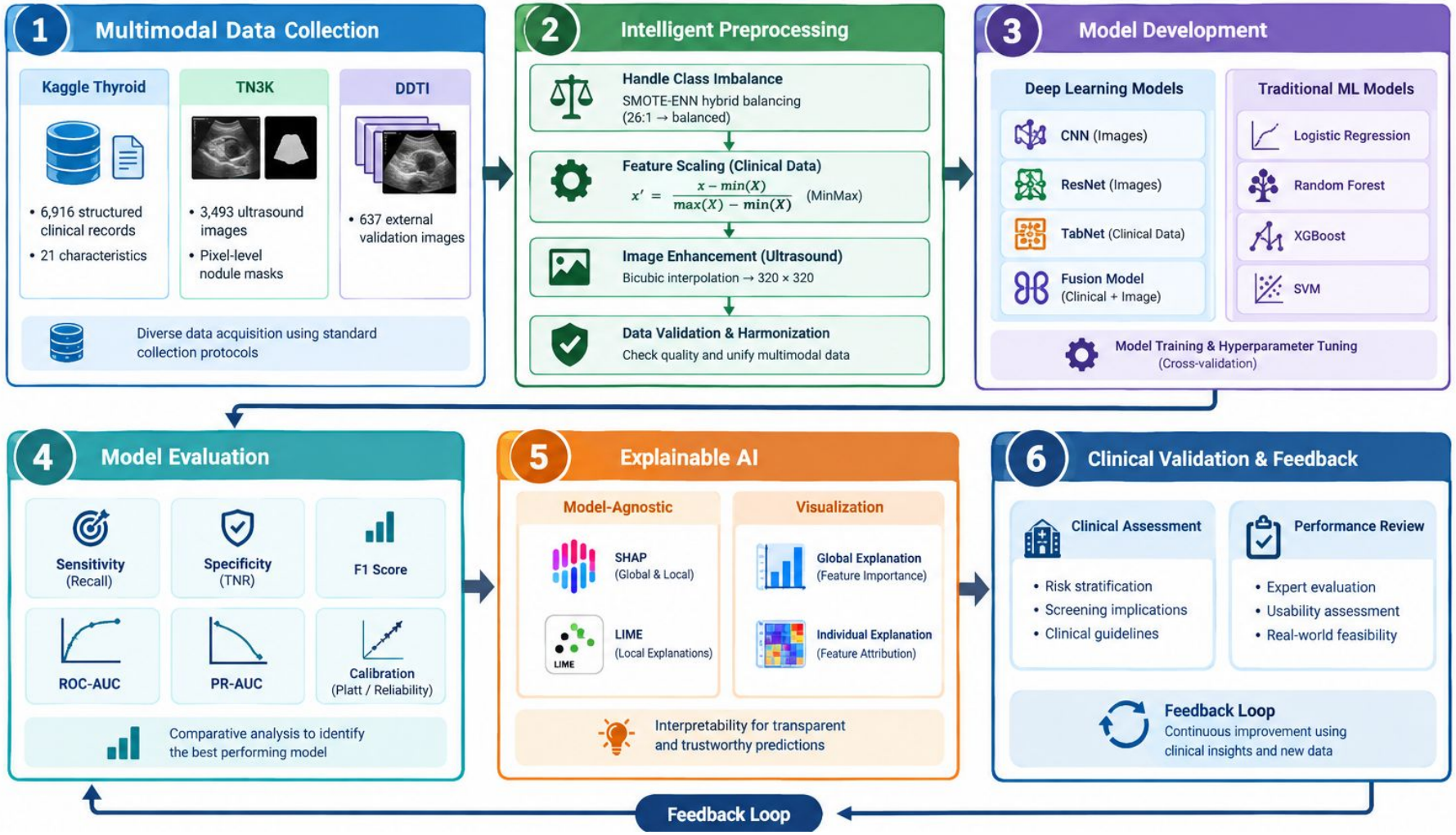
Imputation, IQR-based outlier handling, Min–Max scaling, categorical encoding.

Image preprocessing

Z-score normalization, Lee filtering, 320 × 320 resizing, augmentation.

Clinical features include biochemical and demographic variables; the top-ranked features include TSH, T3, FTI, sex, and age.

Proposed Methodology



Evaluation: stratified 5 × 2 nested cross-validation for clinical models; TN3K internal and DDTI external ultrasound evaluation.

Methodology I — Clinical Branch

1. Imbalance handling

SMOTE with 5 neighbors + ENN with 3 neighbors; noisy samples are removed.

2. Feature ranking

Weighted score combines ANOVA (0.4), information gain (0.3), and RF importance (0.3).

3. Hybrid prediction

Confidence-weighted combination of Random Forest and DNN probabilities.

Clinical training configuration

01. Random Forest: 200 trees; mtry = 4.

02. DNN: three hidden layers, ReLU activation, 0.3 dropout.

03. Clinical evaluation: stratified 5 × 2 nested cross-validation; seed = 42.

04. Statistical analysis: Mann–Whitney U test, Cohen's d, and 95% bootstrap confidence intervals.

Methodology II — Ultrasound Branch

FCN–TransUNet

A. FCN extracts local spatial features.

B. TransUNet models global contextual information.

C. Multi-scale features are fused and decoded into a segmentation mask.

D. Input images resized to 320 × 320 with augmentation.

Composite loss

Weighted objective

$$L = 0.4\,BCE + 0.3\,Dice + 0.2\,Boundary + 0.1\,Focal$$

Explainability

A. SHAP: global and local clinical feature contributions.

B. LIME: local clinical prediction explanations.

C. Grad-CAM: image regions influencing model output.

DDTI is used as an independent dataset to assess cross-dataset generalization.

Results I — Clinical Classification

0.9798
RF accuracy

0.9959
RF AUC

0.9762
RF+DNN accuracy

0.7179
RF+DNN F1-score

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.9798	0.6486	0.9600	0.7742	0.9959
DNN	0.8526	0.1402	0.6000	0.2273	0.8462
RF + DNN	0.9762	0.6269	0.8400	0.7179	0.9803

Interpretation: RF had the strongest discrimination; RF–DNN offered a combined precision–recall profile. Accuracy alone is insufficient under class imbalance.

Results II — Ultrasound Segmentation

0.9942
TN3K AUC

0.8974
TN3K Dice

0.8138
TN3K IoU

0.9766
DDTI AUC

Model	AUC	F1	IoU	Dice	HD95 (mm)
FCN + TransUNet	0.9942	0.8974	0.8138	0.8974	16.87
FCN + CPFNet	0.9932	0.8902	0.8022	0.8902	20.98

Internal evaluation

Strong overlap and discrimination on TN3K.

External evaluation

DDTI AUC = 0.9766; performance remains promising across datasets.

Discussion

Imbalance-aware learning

SMOTE-ENN aims to improve minority-class learning while removing noisy samples.

Hybrid architectures

RF–DNN combines interpretable tree-based patterns with nonlinear DNN representations.

Cross-dataset testing

DDTI evaluation provides evidence of generalization beyond TN3K, but not clinical validation.

Statistical evidence reported in the paper

Comparison	p-value	Cohen's d
RF+DNN vs RF (accuracy)	3.2×10^{-7}	1.24
FCN+TransUNet vs U-Net (Dice)	8.4×10^{-6}	1.56
FCN+TransUNet vs DeepLabV3+ (Dice)	2.3×10^{-6}	1.42
TN3K vs DDTI (AUC/generalization)	2.1×10^{-5}	0.89

Caution: reported results are experimental evidence; further independent, multi-center and expert-led validation is needed.

Conclusion & Future Work

CONCLUSION

1) ThyroNet-XAI unifies clinical prediction and ultrasound segmentation.

2) SMOTE-ENN and feature ranking support the clinical pipeline.

3) FCN–TransUNet achieved Dice 0.8974 on TN3K.

4) SHAP, LIME, and Grad-CAM add complementary interpretability.

FUTURE WORK

1. Multi-center validation with diverse patient populations.

2. More rigorous prospective clinical evaluation.

3. Improve robustness to scanner and acquisition differences.

4. Optimize computational efficiency and deployment readiness.

ThyroNet-XAI is a research decision-support framework—not a replacement for clinical expertise.

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Questions & Discussion

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Thank You

Keep Learning, Keep Grinding